Viridis Research — certification record

Run `runs/2026-11-13_P5` 2026-11-13 project P5, task P5-T001 engine `18cc6c42e7b3`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/43335904766`

Two Corrections to Early Intelligence Bound Sources: An Inconsistent Axiom and Finite Compensation Under a Power Ceiling
Justin Hart, Viridis LLC

Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**ASSUMED**	conjecture: The unrestricted historical Intelligence Bound axiom (rate quantified over all reals, guards kB>0, T>0, P>=0, 0<=D<=1) is false as written (proposed statement P5.UnrestrictedBoundFalse).
C2	**ASSUMED**	conjecture: For cap P D C = P*D/C with C>0, P>0, 0<dD<D: the power P*D/(D-dD) is the unique power restoring capacity after the loss; under a ceiling P<=Pmax it is attainable iff dD <= D*(1-P/Pmax); it diverges as dD -> D from below; and at D=0 no power restores a positive capacity (proposed statement P5.CompensationUnderCeiling).

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 3 entries, 3 DOIs, 3 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 2.
- P5.UnrestrictedBoundFalse: compiled; no probe tactic closes it; submitted for sealing
- P5.CompensationUnderCeiling: compiled; no probe tactic closes it; submitted for sealing

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED**: the verifier re-ran the claim's analysis script in a sandbox with no network, on input files pinned by hash, and reproduced the cited result files; this does not certify where the data came from or the inference drawn from it. **REPLICATED**: not yet enabled. **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN or TESTED are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

Two Corrections to Early Intelligence Bound Sources: An Inconsistent Axiom and Finite Compensation Under a Power Ceiling

Justin Hart
Viridis LLC

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Abstract

Two declarations in early Viridis Lean sources claim more in their names and comments than their formal text supports. First, an axiom of January 2026 states an "Intelligence Bound" $\dot{I} \leq \frac{P}{k_B T \ln 2} D$ for every real rate $\dot{I}$, with no hypothesis linking the rate to any system; the instance $\dot{I} = 1$, $P = D = 0$, $k_B = T = 1$ satisfies every guard and gives $1 \leq 0$, so the axiom is false and any environment assuming it is inconsistent. Second, a theorem named `no_finite_compensation` in fact proves that the finite power $PD/(D - \Delta D)$ exactly compensates a partial loss ΔD of the diversity factor D. We separate four facts about the capacity algebra cap $= PD/C$ that the surrounding prose mixes: exact and unique compensation in the interior, the condition $\Delta D \leq D(1 - P/P_{\max})$ under a power ceiling, divergence of the required power as $\Delta D \to D$, and impossibility at $D = 0$. We write both corrections as Lean 4 statement modules over Mathlib, with sanity checks and negative controls, and propose them for sealing. The mathematics is elementary. Until the statements are sealed and proved, every claim in this note is a conjecture. Nothing here concerns whether any agent preserves biodiversity, nor the Intelligence Bound as a physical conjecture.

1 Introduction

Machine-checked statements are only as good as the match between their formal text and the claims made about them. A review of the Viridis research corpus on 11 November 2026 found two early Lean declarations where the two diverge. Both were found by reading the sources; neither had been checked by the Lean kernel. Project P5 checks them in the open and replaces the prose with exact statements. This note reports the first step: two proposed statements, written so that a later session can prove them against a fixed, sealed target.

The question matters because the historical prose was used to motivate a stronger argument, that capacity lost with biodiversity cannot be bought back with power. If an axiom is inconsistent, everything derived in its presence is vacuous; if a theorem proves the opposite of its name, citing it supports the wrong conclusion. Correcting both is a matter of record keeping, not new mathematics.

2 Background

Landauer's principle [2] associates a minimal dissipation of order $k_B T \ln 2$ with each logically irreversible bit operation. The early sources define, over the reals,

$$\text{landauerEnergy}(k_B, T) = k_B T \ln 2, \qquad \text{maxBitRate}(P, k_B, T) = \frac{P}{k_B T \ln 2},$$

and a capacity $P \cdot D$ scaled by a cost, with $D \in [0, 1]$ a diversity factor. We use Lean 4 [1] and Mathlib [3]. The two historical files are identified by SHA-256 prefixes (`37c017145dcf...2098c4` and `8b1194634...bee273`); they are quoted, not imported or copied.

3 Finding 1: the unrestricted axiom is false

The historical axiom reads

$$\forall\, \dot{I}, P, D, k_B, T \in \mathbb{R} :\ k_B > 0,\ T > 0,\ P \geq 0,\ 0 \leq D \leq 1 \implies \dot{I} \leq \text{maxBitRate}(P, k_B, T)\, D.$$

Conjecture 1 (`P5.UnrestrictedBoundFalse`)**.** *The displayed proposition is false.*

The witness $\dot{I} = 1$, $P = D = 0$, $k_B = T = 1$ meets every guard, and the right-hand side is 0. The rate $\dot{I}$ is an unconstrained real, so no physics enters. As a negative control, adding the hypothesis $\dot{I} \leq \text{maxBitRate}(P, k_B, T)\, D$ makes the implication trivially true; the missing ingredient is a definition of the rate from a modelled system. This says nothing about a physical Intelligence Bound, which would need such a model.

4 Finding 2: finite compensation exists

Let $\text{cap}(P, D, C) = PD/C$. The theorem `no_finite_compensation` proves, for $P > 0$ and $0 < \Delta D < D$, that $\text{cap}\big(P + \frac{P\Delta D}{D - \Delta D},\, D - \Delta D\big) = \text{cap}(P, D)$; since $P + \frac{P\Delta D}{D-\Delta D} = \frac{PD}{D-\Delta D}$, a finite power restores the capacity. We separate four facts.

Conjecture 2 (`P5.CompensationUnderCeiling`)**.** *Let $C > 0$ and $P > 0$.*

1. Interior. *If $0 < \Delta D < D$, then $P' = PD/(D - \Delta D)$ satisfies $\text{cap}(P', D - \Delta D, C) = \text{cap}(P, D, C)$, and it is the only such P'.*
2. Ceiling. *If moreover $P \leq P_{\max}$, some $P' \leq P_{\max}$ restores the capacity if and only if $\Delta D \leq D\,(1 - P/P_{\max})$.*
3. Divergence. *If $D > 0$, then $PD/(D - \Delta D) \to \infty$ as $\Delta D \to D^-$.*
4. Complete loss. *If $D > 0$, then $\text{cap}(P', 0, C) = 0 < \text{cap}(P, D, C)$ for every P'.*

For part 2: with $D - \Delta D > 0$ and $P_{\max} \geq P > 0$, $PD/(D - \Delta D) \leq P_{\max} \iff PD \leq P_{\max}(D - \Delta D) \iff \Delta D \leq D - PD/P_{\max}$. Table 1 lists the sanity checks included in the module.

Check	Instance	Expected	Tactic
Witness	$P = D = C = 1$, $\Delta D = \frac{1}{2}$	$P' = 2$	`norm_num`, `decide`
Ceiling met	$P_{\max} = 2$	$\frac{1}{2} = 1 \cdot (1 - \frac{1}{2})$	`norm_num`, `decide`
Ceiling missed	$P_{\max} = \frac{3}{2}$	$\frac{1}{2} > \frac{1}{3}$	`norm_num`, `decide`
Complete loss	$P' = 1000$, $D = 0$	$0 < 1$	`norm_num`
Drop $P \le P_{\max}$	$P_{\max} = -1$	RHS true, LHS false	`linarith`
Drop $\Delta D < D$	$\Delta D = 2$, $P_{\max} = 1$	LHS true ($P' = -1$), RHS false	`norm_num`

Table 1: Sanity checks and negative controls in `P5.CompensationUnderCeiling`. The `decide` checks use cleared-denominator forms over $\mathbb{Z}$.

5 Formal development

Both statements are self-contained Lean modules importing only Mathlib; all definitions live inside the statement's namespace and no axiom is declared. The core of the second module is:

```
def cap (P D C : ℝ) : ℝ:= P * D / C

def Statement : Prop :=
  (∀ P D dD C : ℝ, 0 < C →0 < P →0 < dD →dD < D →
    cap (P * D / (D - dD)) (D - dD) C = cap P D C ∧
    ∀P' : ℝ, cap P' (D - dD) C = cap P D C →P' = P * D / (D - dD)) ∧
  (∀ P D dD C Pmax : ℝ, 0 < C →0 < P →0 < dD →dD < D →P ≤Pmax →
    ((∃ P' : ℝ, P' ≤Pmax ∧cap P' (D - dD) C = cap P D C) ↔
      dD ≤D * (1 - P / Pmax))) ∧
  (∀ P D C : ℝ, 0 < C →0 < P →0 < D →
    ∀M : ℝ, ∃δ: ℝ, 0 < δ∧∀dD : ℝ, D - δ< dD →dD < D →
      M < P * D / (D - dD)) ∧
  (∀ P D C : ℝ, 0 < C →0 < P →0 < D →
    ∀P' : ℝ, cap P' 0 C = 0 ∧cap P' 0 C < cap P D C)
```

and of the first:

```
def landauerEnergy (kB T : ℝ) : ℝ:= kB * T * Real.log 2
def maxBitRate (P kB T : ℝ) : ℝ:= P / landauerEnergy kB T

def Statement : Prop :=
  ¬(∀ dIdt P D kB T : ℝ,
      kB > 0 →T > 0 →P ≥0 →D ≥0 →D ≤1 →
      dIdt ≤maxBitRate P kB T * D)
```

Divergence is written with an explicit δ rather than a filter limit, so the statement can be read without Mathlib's filter vocabulary.

6 Results

Both modules compile against the pinned toolchain, and all of their sanity checks and negative controls compile. We observe two points about the specification that we passed to the rationale files. (i) A negative control for dropping $\Delta D < D$ needs $\Delta D > D$: at $\Delta D = D$ both sides of part 2 are false, so the equivalence still holds there. (ii) A negative control for dropping $P \le P_{\max}$ needs $P_{\max} \le 0$: for $0 < P_{\max} < P$ both sides are false. Neither point changes the proposed statements.

Real-number arithmetic is not decidable and `decide` on $\mathbb{Q}$ got stuck in kernel reduction, so the `decide` checks use equivalent integer forms and the real checks use `norm_num`.

7 Limitations

The results are elementary corrections of record, not new theorems. They say nothing about whether any agent preserves biodiversity: a capacity upper bound does not say the bound is attained, that capacity serves an agent's goal, that biodiversity is the only source of the relevant information, or that preservation is the best action once costs are counted. They say nothing about the Intelligence Bound as a physical conjecture, and cap is an algebraic model, not a description of a physical system. The claim that no other corpus file uses the axiom comes from reading the corpus and is not checked by these statements. No proof is claimed in this note; the statements are conjectures until sealed and proved.

8 Reproducibility

The modules are

- `proposals/P5.UnrestrictedBoundFalse.lean` and
- `proposals/P5.CompensationUnderCeiling.lean`,

with rationale files alongside. They compile with `lean` from toolchain `leanprover/lean4:v4.28.0` and Mathlib commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. No data or external code is used.

9 Changelog

- 2026-11-13 (P5-T001): first version. Proposed both statements with sanity checks and negative controls; noted the conditions the negative controls need.

References

[1] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

[2] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

[3] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. Association for Computing Machinery, 2020. doi: 10.1145/3372885.3373824.